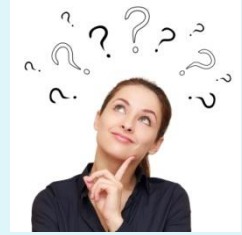


RESEACH METHODOLOGY ETHICAL ISSUES IT498



What is this course about???



- ▶ Preparing you to write professional **bachelor project report**. (**Objective 1**)
 - Your bachelor report is sometimes called “undergraduate research report” 😊

Therefore!!!

- ▶ To learn and conduct basic research in IT domain. (**Objective 2**)



Textbooks and Course Material

- ▶ There is no compulsory textbook for this course. However the following sources can be suggested for reading.
 - C. R. Cothari, Research Methodology Methods and Techniques, Second Edition, New Age International Publishers. Paul D. Leedy, Jeanne Ellis Ormrod, Practical Research: Planning and Design, 10/E, Pearson Educational International: New Jersey

Demo Report

- ▶ Each student is expected to define a problem (**preferably your bachelor project topic**), prepare a report containing **all the report structural items** and do a 10 min **oral presentation** about the topic. (**Basic**)
- ▶ By successfully completing this assignment the students will improve their skills in doing literature search and scientific communication.
- ▶ The students are encouraged to communicate with the **supervisor**. The supervisor can also adjust **40% marks** on this report if he wishes so.

Project Planning

Learning Objectives

- ▶ Develop a working knowledge of basic project planning and scheduling skills
- ▶ Learn how to use several basic tools
 - Work breakdown structures
 - Resource allocation
 - Gantt charts

What is a Project?

- ▶ A project has a single objective that must be accomplished through the completion of tasks that are unique and interrelated
- ▶ Projects are completed through the deployment of resources
- ▶ Projects have **scopes**, **schedules**, and **costs** and are accomplished within specific **deadlines**, **budgets**, and according to **specification**

Project Planning Activity

Identify a class project to use as we move through this lesson

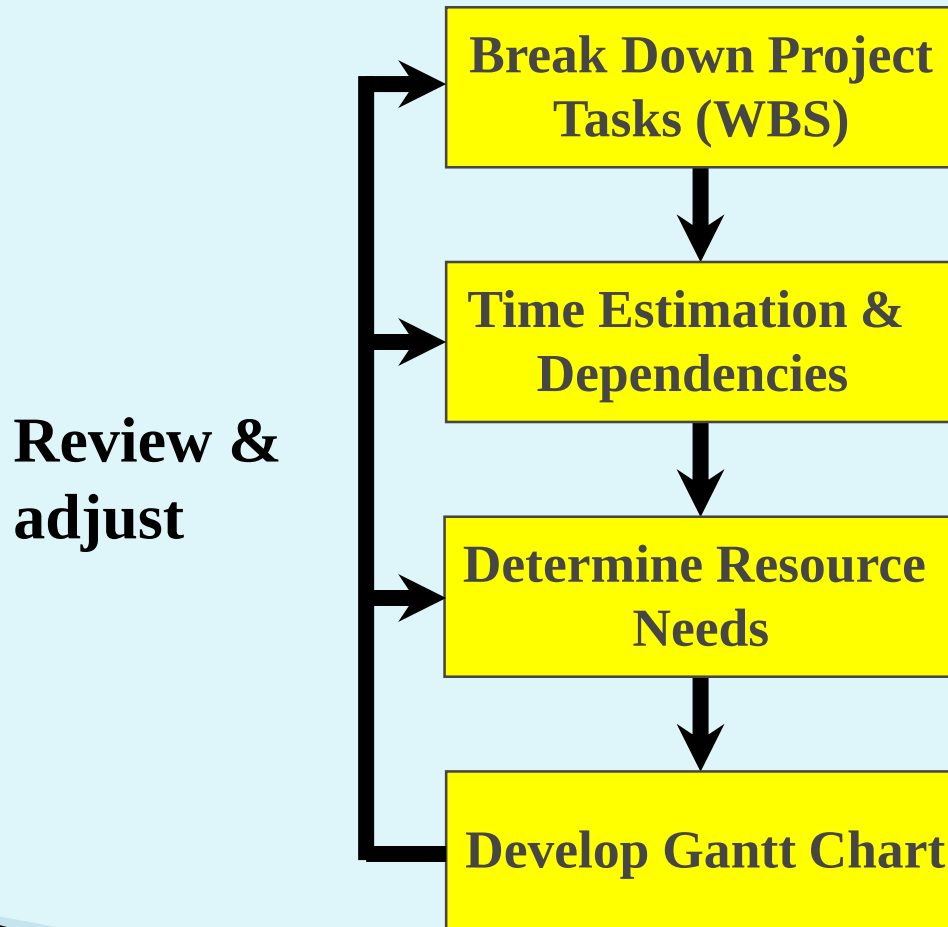
- What is your project name?
- What is the main goal of your team's project?



Phases of Project Management

- Define the project's scope
- Develop the project's plan
- Implement the plan & control the process
- Close out the project

Develop the Project's Plan



What is a Work Breakdown Structure (WBS)?

- ▶ A hierarchical representation of activities
- ▶ It starts with the major project areas to be accomplished
- ▶ It breaks the project areas into actionable pieces of work, segmenting elements into appropriate sublevels

WBS – Activity Levels

- ▶ Level 1 – Identify major objective areas or categories
- ▶ Level 2 – Begin to divide the areas into sub-tasks
- ▶ Level 3 – Continue to break down the sub-tasks into actionable items
- ▶ The lowest level associated with a branch in the hierarchy is referred to as a “work package”

of levels depends on project complexity

How much detail do you need?

- ▶ Does the WBS contain enough detail to evaluate progress?
 - Do you have clear accountability for each work package?
 - Are there start & end events?
 - Can you easily estimate time & cost?
 - Is there a clearly defined deliverable?

Project Planning Activity

- ▶ Create a work breakdown structure for your project



Time Duration

Managing a project requires awareness of two time frames

1. The amount of effort a task will take (in time), e.g., 3 hours to write a report
2. The calendar span over which the activity will occur, e.g., the report will be done within a week

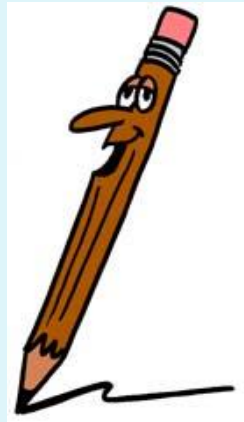


Uncertainty in Time Estimates

- ▶ Some activities will take longer and others will go faster than expected
 - Sources of uncertainty:
 - Varying knowledge and skills
 - Individual difference in approaching work
 - Mistakes or misunderstandings
 - *Unexpected events (!!)*

Project Planning Activity

- ▶ For each work package, estimate the time duration in days
- ▶ Determine the time duration of each higher level activity
- ▶ Repeat !!!



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Identifying Resources

To accomplish each activity identified in the WBS, requires the use of resources:

- Personnel (who, how many, their skills)
- Space (meeting location, lab facilities)
- Equipment (rent, own, purchase, how long needed)
- Money (budget limitations)



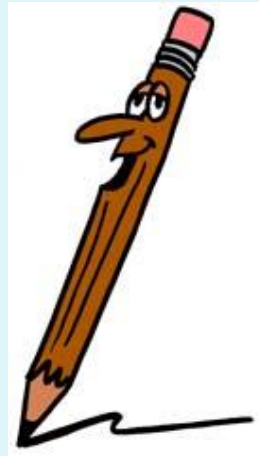
Assigning Resources

- ▶ Assign resources to the appropriate tasks (personnel, space, equipment, money)
- ▶ Be realistic– no one can be in two places at one time



Project Planning Activity

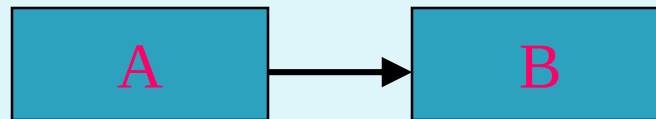
- ▶ What are the resources needed for each task in the WBS of your bachelor project?



Dependencies

Dependencies are the relationships between activities

“Finish to Start” example



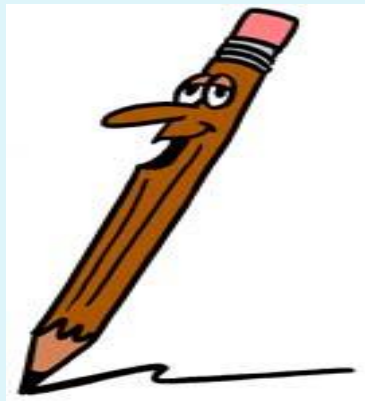
Predecessor Task: A

Successor Task: B

Arrow head indicates dependency relationship:
Task B cannot begin until Task A is complete

Project Planning Activity

- ▶ Number all tasks in your WBS
- ▶ Identify the dependencies between each task



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The Gantt Chart: A Visual Scheduling Tool

- ▶ Graphically represents WBS information
- ▶ Shows dependencies between tasks, time duration, personnel, and other resource allocations
- ▶ Tracks progress towards project completion

Building a Gantt Chart – Axes

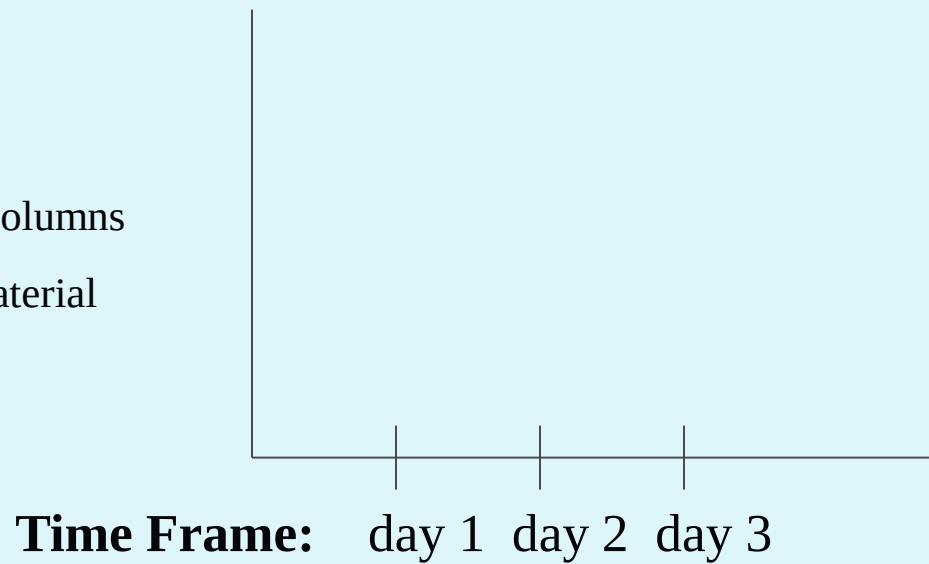
- ▶ List all tasks and milestones from the WBS along the vertical axis
- ▶ List time frame along the horizontal/vertical axis

Tasks:

Design support columns

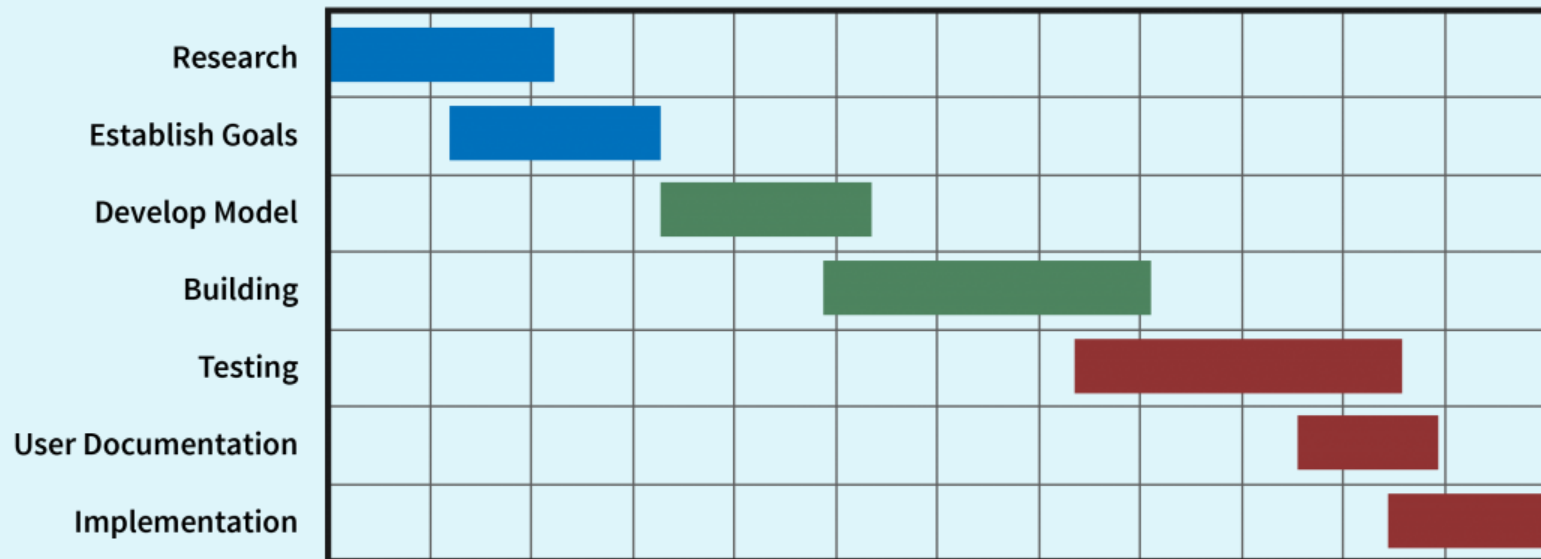
Select final material

Design...



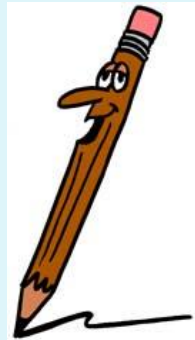
Building a Gantt Chart—Dependencies

Gantt Chart



Assignment No 1: Project Planning Activity

- ▶ Based on the WBS (tasks, durations, and dependencies), create a Gantt Chart for your project. (Only approximate)



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Next Step: Implementing the Plan

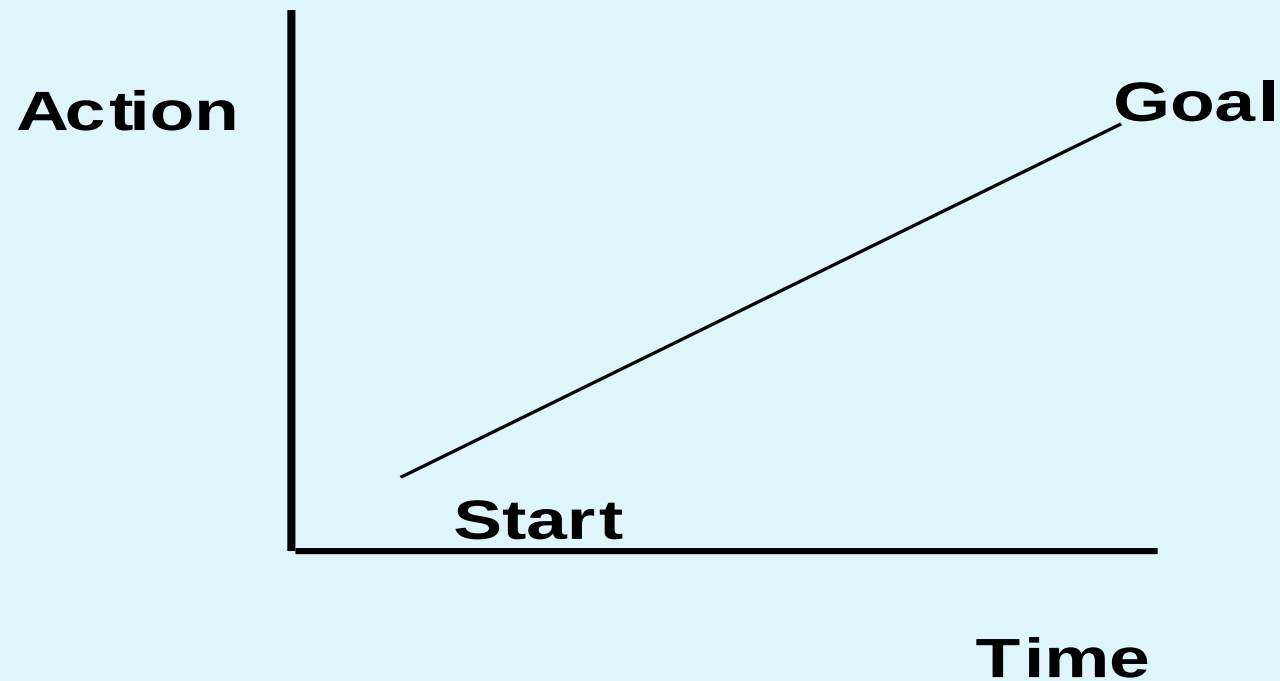
- ▶ After the plan is complete, everyone should know who is responsible for each activity and when each task must be complete to ensure project success.

Responsibility Matrix

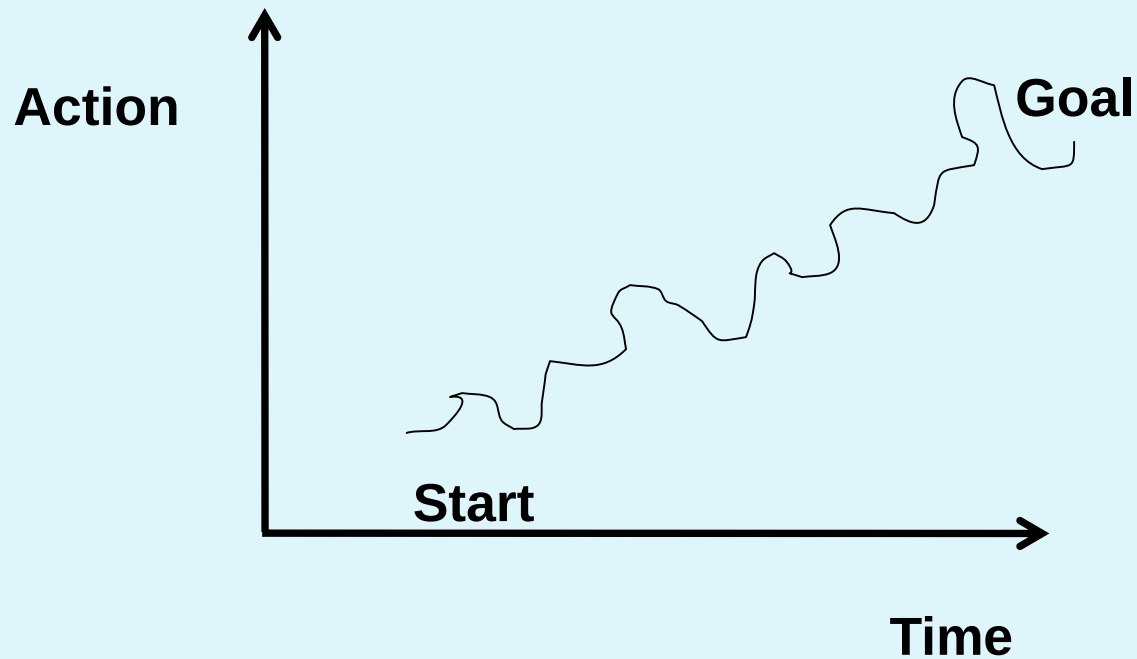
Creates accountability by assigning each task to a person

Task	Joe	Mary	Renee
Activity 1		x	
Activity 2	x		
Activity 3			x

Managing Your Projects: Time



In Reality...



Methods to Track Time

- ▶ Time management books or schedulers
- ▶ PDAs
- ▶ A piece of paper to write things down
- ▶ Memory

